MIL-S-43012C(ME)
16 June 1988
SUPERSEDING
MIL-S-43012B(ME)
24 August 1981

MILITARY SPECIFICATION

SAMPLER, SOIL

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an instrument used to take soil samples.

1.2 Part numbering system. Items procured under this specification shall be identified by the following specification-based part number: M43012-1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and drawings. The following specifications, standards, and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

2.1.2 Other Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issued shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13200E9850	- Sampler, Soil.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1	- Surface Texture.

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ANSI/SAE J429 - Mechanical and Material Requirement for Externally Threaded Fasteners.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Sheet, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The soil sampler shall be in accordance with TA13200E9850, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the soil sampler. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the sampler from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish samplers conforming to this specification.

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3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 4.1.2).

3.3.1 Material deterioration prevention and control. The soil sampler shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the soil sampler may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the soil sampler may be newly fabricated from recovered materials to the maximum extent practicable, provided the soil sampler produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the soil sampler.

3.4 Dimensions. Tolerance and gages for metal fits shall conform to the limits specified herein and on the applicable drawings.

3.5 Finish. Surface finishes shown on the drawings shall conform to ANSI/ASME B46.1.

3.6 Identification marking. The soil sampler shall be identified in accordance with MIL-STD-130, and as specified herein.

3.7 Workmanship.

3.7.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners,

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except when sharp edges or corners are required or where they are not detrimental to safety.

3.7.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to ANSI/SAE J429 or capscrews shall be of sufficient strength to withstand the tensile of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.7.3 Aluminum welding. Aluminum parts to be welded shall be sheared or sawed. Flame cutting shall not be used. The surface of aluminum parts to be welded shall be free from paint, grease, oxide film and all other foreign matter, which shall be removed using a stainless steel wire brush. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed or chipped. Welds shall not be hammered. Completed welds shall be free from discontinuities such as base metal cracks adjacent or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit or splatter.

3.7.4 Machine work. Tolerance and gage for metal fits shall conform to the limits specified herein, and on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality assurance program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirement of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Acceptable Quality Level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The soil sampler shall be examined as specified herein for the following defects.

Major

101. Materials not as specified (see 3.3).
102. Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finishes or treatment (see 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces or parts incorporated in the soil sampler (see 3.3.2).
106. Dimensions not as specified (see 3.4).

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107. Components missing or not as specified.
108. Workmanship not as specified (see 3.7).
109. Metal fabrication not as specified (see 3.7.1).
110. Bolted connections not as specified (see 3.7.2).
111. Aluminum welding not as specified (see 3.7.3).
112. Machine work not as specified (see. 3.7.4).

Minor

201. Finish not as specified (see 3.5).
202. Identification marking missing, incomplete or illegible (see 3.6).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

113. Containers not as specified. Each incorrect container shall be considered one defect.
114. Handles not removed as specified for level A (see 5.1.1).
115. Fiberboard boxes not closed as specified for level A or commercial (see 5.2).
116. Strapping not as specified for level A or B (see 5.2.1 and 5.2.2).
117. Marking illegible, incorrect or missing (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The handles of the soil sampler shall be removed to effect a reduction in cube. Each sampler with removed handles shall be placed in a fiberboard box conforming to PPP-B-636. The box shall be closed in accordance with method V of the appendix to the box specification. The box contents shall be cushioned in accordance with MIL-STD-1186 to prevent movement and damage.

5.1.2 Commercial. The soil samplers shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, level B or commercial, as specified (see 6.2).

5.2.1 Level A. The preserved samplers shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional, grade B or PPP-B-621, class 2, style optional. The maximum weight of the contents shall not exceed 200 pounds. Strapping of the boxes shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level B. The preserved sampler shall be packed as specified in 5.2.1 for level A, except that the boxes shall be domestic type and strapping shall be finish A. As an alternate, boxes shall conform to PPP-B-636, V3c or greater, and closed in accordance with method V of the appendix to the box specification. The weight of the contents of the fiberboard box shall not exceed 200 pounds.

5.2.3 Commercial. Commercial packing shall be in individual containers in accordance with ASTM D 3951.

5.3 Marking. Marking for level A or B shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with ASTM D 3951; each container shall be marked with cube and gross weight.

6. NOTES

6.1 Intended use. The soil samplers are intended to take soil samples and eject the samples directly into the remolding cyclinder.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Part number (M43012-1).
c. Timeframe required for submission of first article and the number of units required (see 3.2).
d. When a first article is not required (see 3.2).
e. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

Sampler, soil
Soil, sampler

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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Review activity:
DLA - GS

Project 6635-A134

User activity:
Army - GL

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43012C(ME)	Sampler, Soil

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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